



Reflect with us - Week 7



This assignment will not be graded and is only for practice.

1 point

Let L_1 and L_2 be affine subspaces of R^3 , where $L_1 = U$ and $L_2 = (2, 0, 1) + W$, for some vector subspaces U and W of R^3 . Let a basis for U be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ and a basis for W be given by $\{(1, 0, 1), (0, 1, 1)\}$. Suppose there is a linear transformation $T: U \rightarrow W$ such that $(0, 1, 0) \in \ker(T)$, $T(2, 0, 1) = (0, 1, 1)$ and $T(1, 1, 0) = (1, 0, 1)$. An affine mapping $f: L_1 \rightarrow L_2$ is obtained by defining $f(u) = (2, 0, 1) + T(u)$, for all $u \in U$. Which of the following options are true?

☐

$$L_1 = R^3$$

☐

$$L_2 = \{(x, y, z) \mid x + y - z = 1\}$$

☐

$$L_2 = R^3$$

☐

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

☐

$$f(x, y, z) = (x - 2z, z, x - z)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$L_1 = R^3$$

$$L_2 = \{(x, y, z) \mid x + y - z = 1\}$$

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

Solution:

Discussion on Option 1: It is given that $L_1 = U$ and $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ is a basis of the vector subspace U .

The dimension of U is

Hint

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

$L_1 = U$ is a subspace of R^3 . To know whether $L_1 = U$ is a proper subspace



R^3 or not, we just have to compare their dimensions.

1 point

Now, can you see why $L_1 = U = R^3$ $L_1 = U = R^3$?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

Yes

Discussion on Options 2 and 3: Discussion on Options 2 and 3: It is given that $L_2 = (2, 0, 1) + W$ $L_2 = (2, 0, 1) + W$ and $\{(1, 0, 1), (0, 1, 1)\}$ $\{(1, 0, 1), (0, 1, 1)\}$ is a basis for the vector subspace W . Now to find the explicit description of L_2 , we have to find the explicit description of W first.

$$W = \{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in R\} \\ = \{(a, b, a + b) \mid a, b \in R\}$$

$$\{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in R\} = \{(a, b, a + b) \mid a, b \in R\}$$

As W is a subspace of R^3 , we want to express it using the usual x, y, z co-ordinate and the relation between them. From the above description we have, $x = a$, $y = b$, $z = a + b$, which implies that $z = x + y$. So

$$W = \{(x, y, z) \mid x + y - z = 0, \text{ where } x, y, z \in R\} \\ = \{(x, y, z) \mid x + y - z = 0, \text{ where } x, y, z \in R\}$$

1 point

Now, are you able to find the equation of L_2 ?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

Yes

1 point

Do you agree that the sets $\{(2 + x, y, z + 1) \mid x + y - z = 0\}$ and $\{(a, b, c) \mid a + b - c = 1\}$ are the same?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

Yes

Discussion on Option 4 and 5: Discussion on Option 4 and 5:

Step 1: Step 1: Here, we have to find an algebraic expression for the affine map $f: L_1 \rightarrow L_2$. To get an algebraic expression for f , we have to find the algebraic expression which represents the linear transformation $T: U \rightarrow W$.

Given:

$$T(0, 1, 0) = (0, 0, 0), \quad T(2, 0, 1) = (0, 1, 1), \quad T(1, 1, 0) = (1, 0, 1)$$

1 point

Is T uniquely determined by the above information?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

Yes

Step 2: Step 2: Since $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$ is a basis of $U = \mathbb{R}^3$, we can write any element $(x, y, z) \in U$ as a linear combination of $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$.

$$(1) \quad (x, y, z) = a(0, 1, 0) + b(2, 0, 1) + c(1, 1, 0)$$

Apply T on both sides:

$$T(x, y, z) = aT(0, 1, 0) + bT(2, 0, 1) + cT(1, 1, 0)$$

$$T(x, y, z) = a(0, 0, 0) + b(0, 1, 1) + c(1, 0, 1)$$

$$T(x, y, z) = a(0, 0, 0) + b(0, 1, 1) + c(1, 0, 1)$$

To get an algebraic expression for $T(x, y, z)$, we just have to know the values of a , b and c in terms of x , y and z .

1 point

Now, can you find the values of a , b and c using equation (1)?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

Yes

Solve the above system for a , b and c .

1 point

What is the value of a ?

☐

z

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$y + 2z - xy + 2z - x$

1 point

What is the value of bb ?

☐

zz

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

zz

1 point

What is the value of cc ?

☐

zz

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x - 2zx - 2z$

Step 3:Step 3:

$$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z) \quad T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$$

By definition:

$$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U. \quad f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U.$$

By substituting the value $T(x, y, z)$ in the above expression, we get

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all } (x, y, z) \in U. f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all } (x, y, z) \in U.$$

[Check Answers](#)